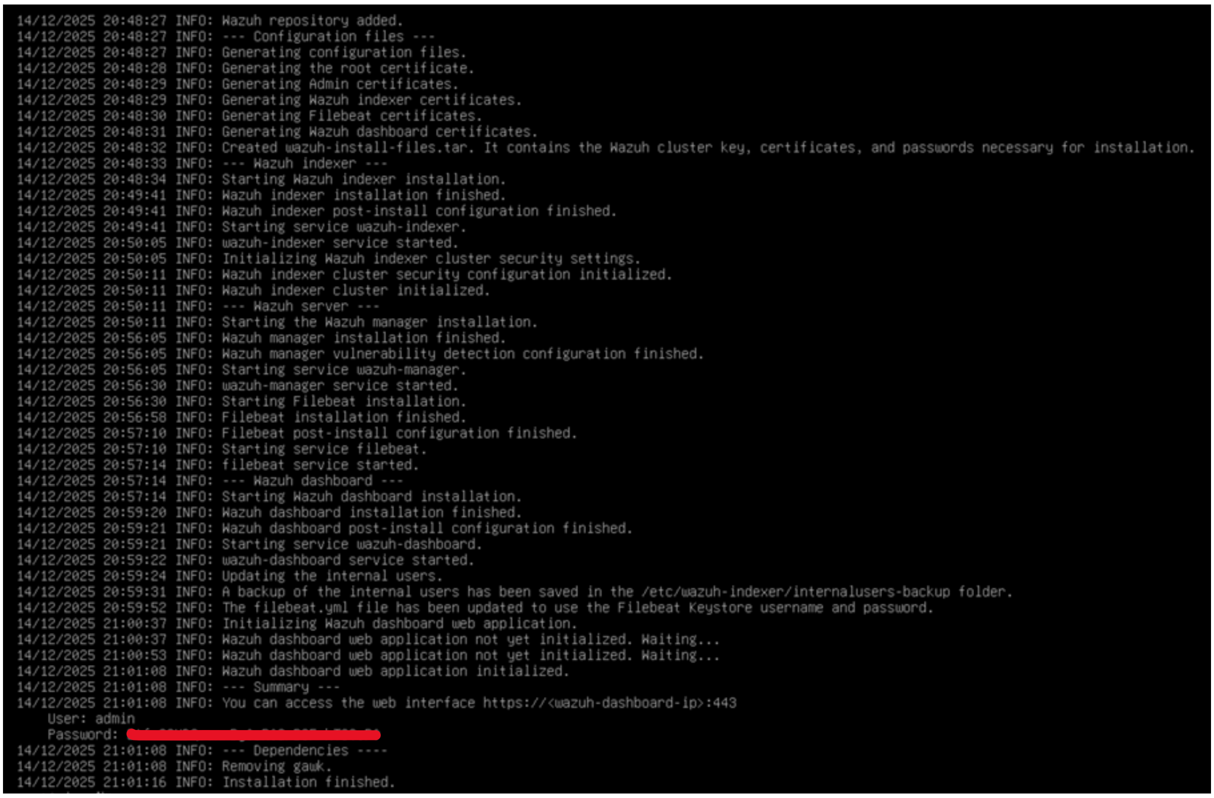
Téléchargement et installation :

apt install curl
apt-get update && apt-get clean && apt-get install -y build-essential
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
apt install -y sudo
sudo bash ./wazuh-install.sh -a



```
14/12/2025 20:48:27 INFO: Wazuh repository added.
14/12/2025 20:48:27 INFO: --- Configuration files ---
14/12/2025 20:48:27 INFO: Generating configuration files.
14/12/2025 20:48:28 INFO: Generating the root certificate.
14/12/2025 20:48:29 INFO: Generating Admin certificates.
14/12/2025 20:48:29 INFO: Generating Wazuh indexer certificates.
14/12/2025 20:48:30 INFO: Generating Filebeat certificates.
14/12/2025 20:48:31 INFO: Generating Wazuh dashboard certificates.
14/12/2025 20:48:32 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
14/12/2025 20:48:33 INFO: --- Wazuh indexer ---
14/12/2025 20:48:34 INFO: Starting Wazuh indexer installation.
14/12/2025 20:49:41 INFO: Wazuh indexer installation finished.
14/12/2025 20:49:41 INFO: Wazuh indexer post-install configuration finished.
14/12/2025 20:49:41 INFO: Starting service wazuh-indexer.
14/12/2025 20:50:05 INFO: wazuh-indexer service started.
14/12/2025 20:50:05 INFO: Initializing Wazuh indexer cluster security settings.
14/12/2025 20:50:11 INFO: Wazuh indexer cluster security configuration initialized.
14/12/2025 20:50:11 INFO: Wazuh indexer cluster initialized.
14/12/2025 20:50:11 INFO: --- Wazuh server ---
14/12/2025 20:50:11 INFO: Starting the Wazuh manager installation.
14/12/2025 20:56:05 INFO: Wazuh manager installation finished.
14/12/2025 20:56:05 INFO: Wazuh manager vulnerability detection configuration finished.
14/12/2025 20:56:05 INFO: Starting service wazuh-manager.
14/12/2025 20:56:30 INFO: wazuh-manager service started.
14/12/2025 20:56:30 INFO: Starting Filebeat installation.
14/12/2025 20:56:58 INFO: Filebeat installation finished.
14/12/2025 20:57:10 INFO: Filebeat post-install configuration finished.
14/12/2025 20:57:10 INFO: Starting service filebeat.
14/12/2025 20:57:14 INFO: filebeat service started.
14/12/2025 20:57:14 INFO: --- Wazuh dashboard ---
14/12/2025 20:57:14 INFO: Starting Wazuh dashboard installation.
14/12/2025 20:59:20 INFO: Wazuh dashboard installation finished.
14/12/2025 20:59:21 INFO: Wazuh dashboard post-install configuration finished.
14/12/2025 20:59:21 INFO: Starting service wazuh-dashboard.
14/12/2025 20:59:22 INFO: wazuh-dashboard service started.
14/12/2025 20:59:24 INFO: Updating the internal users.
14/12/2025 20:59:31 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
14/12/2025 20:59:52 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
14/12/2025 21:00:37 INFO: Initializing Wazuh dashboard web application.
14/12/2025 21:00:37 INFO: Wazuh dashboard web application not yet initialized. Waiting...
14/12/2025 21:00:53 INFO: Wazuh dashboard web application not yet initialized. Waiting...
14/12/2025 21:01:08 INFO: Wazuh dashboard web application initialized.
14/12/2025 21:01:08 INFO: --- Summary ---
14/12/2025 21:01:08 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: [redacted]
14/12/2025 21:01:08 INFO: --- Dependencies ----
14/12/2025 21:01:08 INFO: Removing gawk.
14/12/2025 21:01:16 INFO: Installation finished.
```

Configuration

Machines à surveiller => Mise en place du Wazuh Agent (https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-linux.html)

Déploiement de l'agent (sur serveurs et postes) :

apt-get install curl
apt-get install gnupg apt-transport-https
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && chmod 644 /usr/share/keyrings/wazuh.gpg
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | tee -a /etc/apt/sources.list.d/wazuh.list
apt-get update
WAZUH_MANAGER="192.168.20.2" apt-get install wazuh-agent

systemctl daemon-reload
systemctl enable wazuh-agent
systemctl start wazuh-agent

Modification du fichier de configuration de l'agent (en modifiant la ligne server) :

nano /var/ossec/etc/ossec.conf
server : 192.168.20.2

Sur le manager (serveur SIEM), nous pouvons déjà voir la remonté des 3 agents de nos 3 serveurs Linux :

```
root@siem:~# /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: siem (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: shuffle, IP: any, Active
   ID: 002, Name: irplatform, IP: any, Active

List of agentless devices:
```

Il faut régulièrement vérifier qu'aucun n'est marqué disconnected depuis longtemps

Mise en place de l'agent sur la Kali :

```
┌──(kali㉿kali)-[~]
└─$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.3-1_amd64.deb
--2025-12-17 14:47:36--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.3-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.162.38.122, 3.162.38.84, 3.162.38.60, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.162.38.122|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9362524 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.3-1_amd64.deb'

wazuh-agent_4.7.3-1_amd64.deb          100%[=====================================>]   8.93M  7.24MB/s    in 1.2s

2025-12-17 14:47:37 (7.24 MB/s) - 'wazuh-agent_4.7.3-1_amd64.deb' saved [9362524/9362524]

┌──(kali㉿kali)-[~]
└─$ sudo WAZUH_MANAGER="192.168.20.2" dpkg -i wazuh-agent_4.7.3-1_amd64.deb
Selecting previously unselected package wazuh-agent.
(Reading database ... 160226 files and directories currently installed.)
Preparing to unpack wazuh-agent_4.7.3-1_amd64.deb ...
Unpacking wazuh-agent (4.7.3-1) ...
Setting up wazuh-agent (4.7.3-1) ...

┌──(kali㉿kali)-[~]
└─$ sudo systemctl daemon-reexec

┌──(kali㉿kali)-[~]
└─$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start wazuh-agent
```

Déploiement de Sysmon (pour de meilleurs logs) et de l'agent sur la Windows Cliente :

```
PS C:\Users\lade\Downloads\Sysmon> .\sysmon64.exe -i sysmonconfig.xml -accepteula


System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.90
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

```
PS C:\Users\lade\Downloads\Sysmon> Get-WinEvent -FilterHashtable @{LogName="Microsoft-Windows-S
ysmon/Operational"; ID=1} -MaxEvents 5


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                      Id LevelDisplayName Message
-----------                      -- ---------------- -------
12/17/2025 11:13:41 AM            1 Information      Process Create:...
12/17/2025 11:13:40 AM            1 Information      Process Create:...
12/17/2025 11:13:40 AM            1 Information      Process Create:...
12/17/2025 11:13:40 AM            1 Information      Process Create:...
12/17/2025 11:13:40 AM            1 Information      Process Create:...
```

Pour l’agent Windows : https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html

1. Download the Windows installer to start the installation process.

Configuration de celui-ci :

```
C:\Users\lade\Downloads>wazuh-agent-4.14.1-1.msi /q WAZUH_MANAGER="192.168.20.2"
```

```
C:\Users\lade\Downloads>NET START WazuhSvc
The Wazuh service is starting.
The Wazuh service was started successfully.
```

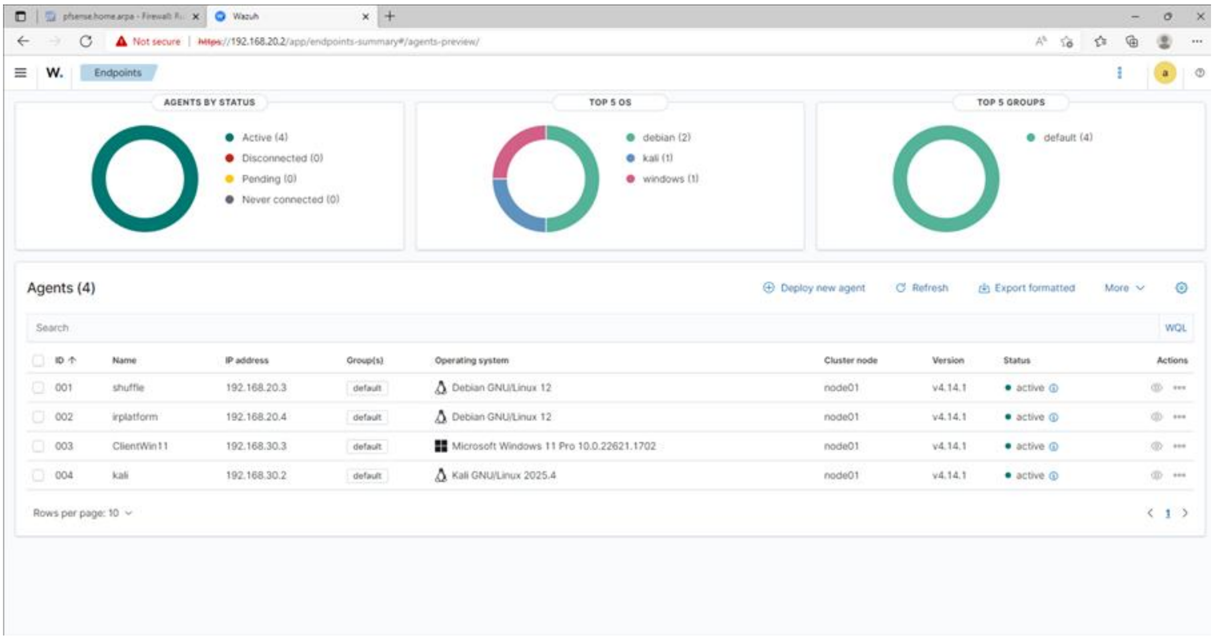
Même fichier sur les windows pour la configuration :

```
    <log_format>eventchannel</log_format>
    <query>Event/System[EventID != 5145 and EventID != 5156 and Ev
5447 and
      EventID != 4656 and EventID != 4658 and EventID != 4663 and
EventID != 4660 and
      EventID != 4670 and EventID != 4690 and EventID != 4703 and
EventID != 4907 and
      EventID != 5152 and EventID != 5157]</query>
  </localfile>

  <localfile>
    <location>System</location>
    <log_format>eventchannel</log_format>
  </localfile>

  <localfile>
    <location>active-response\active-responses.log</location>
    <log_format>syslog</log_format>
  </localfile>

  <localfile>
    <location>Microsoft-Windows-Sysmon/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>
```

Après application des modifications :

```
C:\Users\lade\Downloads>NET STOP WazuhSvc

The Wazuh service was stopped successfully.


C:\Users\lade\Downloads>NET START WazuhSvc

The Wazuh service was started successfully.
```

Toutes nos machines remontent bien sur l’interface principale de Wazuh :



On peut vérifier maintenant les index Wazuh et tester le cluster avec ces commandes :

```
root@siem:~# curl -k -u admin:[redacted] https://localhost:9200/_cat/indices
green open wazuh-states-vulnerabilities-siem                IYFvAnk4TXmbFMVLOyfOvQ 1 0 1388 22   1.2mb   1.2mb
green open wazuh-statistics-2025.50w                        tSyLpORqTJCTetQwB4vEHw 1 0   69  0 149.6kb 149.6kb
green open wazuh-statistics-2025.51w                        mPjXvF40Q4Cp1q9U6wJdnA 1 0  406  0 535.3kb 535.3kb
green open wazuh-states-inventory-packages-siem             Vhb430DxR7GL18kmKbBQMg 1 0  850  0 358.2kb 358.2kb
green open wazuh-states-inventory-hardware-siem             ycnfa5E7RwSK4I_tb-LV0w 1 0    2  0  15.9kb  15.9kb
green open wazuh-states-inventory-ports-siem                -JWM7Kf0Q0KxYXob2sqAng 1 0    4  0  20.2kb  20.2kb
green open wazuh-states-inventory-protocols-siem            XZz8iU-7SmuIaw3cQUsQzA 1 0    2  0  14.6kb  14.6kb
green open wazuh-states-inventory-networks-siem             8DwLnjvjS7i_c6yWvIJkog 1 0    2  0  15.2kb  15.2kb
green open .kibana_1                                        53B1SaBfQECTBL24s83Zkg 1 0    9  1  37.3kb  37.3kb
green open .opendistro_security                             cLqzZv1oQMyiXgOX6y01cQ 1 0   10  0  79.9kb  79.9kb
green open wazuh-alerts-4.x-2025.12.15                      3pBDkpwbSeOx6WyTaJX66A 3 0  470  0   1.5mb   1.5mb
green open wazuh-alerts-4.x-2025.12.14                      IhqOtum-TYeA3T7ABbObfA 3 0  238  0 913.1kb 913.1kb
green open wazuh-states-inventory-users-siem                XV7LXHYcRxiCq31uBOvH1Q 1 0   54  2  89.3kb  89.3kb
green open wazuh-states-inventory-system-siem               P4hyuEpUSyyRrEHJffaL0g 1 0    2  0  19.6kb  19.6kb
green open wazuh-states-inventory-interfaces-siem           rRzg3bKuRfW2KolxwnJ4pQ 1 0    2  0  19.5kb  19.5kb
green open wazuh-states-inventory-browser-extensions-siem SZDjOzhKQUKKTRC19eL_9g 1 0    0  0    208b    208b
green open .plugins-ml-config                               o52xHMOoSiqJLGKd_j0USQ 1 0    1  0     4kb     4kb
green open .opensearch-observability                        Mf-Vrd-KRX-E2he_E2efzg 1 0    0  0    208b    208b
green open wazuh-states-inventory-groups-siem               3jv-vEPhSLOm3F_DP9Y4Dg 1 0  108  0  40.3kb  40.3kb
green open wazuh-monitoring-2025.50w                        2HCK4V1xT_K-QseY-h-jJQ 1 0    0  0    208b    208b
green open wazuh-monitoring-2025.51w                        RUybKbScR9eMdUHjKPdoBg 1 0   20  0 216.3kb 216.3kb
green open wazuh-states-inventory-services-siem             p_2dku6UQyuFhOVrMT-v2g 1 0  225  2 191.1kb 191.1kb
green open wazuh-states-inventory-hotfixes-siem             QmmLAQxAT8m_ZrY1kmZNcA 1 0    0  0    208b    208b
green open wazuh-states-inventory-processes-siem            G4eCYx2pTeWKEDJ931Qf3Q 1 0  408  0 134.2kb 134.2kb
root@siem:~# curl -k -u admin:[redacted] https://localhost:9200/_cluster/health?pretty
{
  "cluster_name" : "wazuh-cluster",
  "status" : "green",
  "timed_out" : false,
  "number_of_nodes" : 1,
  "number_of_data_nodes" : 1,
  "discovered_master" : true,
  "discovered_cluster_manager" : true,
  "active_primary_shards" : 28,
  "active_shards" : 28,
  "relocating_shards" : 0,
  "initializing_shards" : 0,
  "unassigned_shards" : 0,
  "delayed_unassigned_shards" : 0,
  "number_of_pending_tasks" : 0,
  "number_of_in_flight_fetch" : 0,
  "task_max_waiting_in_queue_millis" : 0,
  "active_shards_percent_as_number" : 100.0
}
```

Status green → OK

Vérification des ingestions d’alertes :

```
root@siem:~# curl -k -u admin:[redacted] https://localhost:9200/wazuh-alerts-*/_search?size=5
```


```
{"took":4,"timed_out":false,"_shards":{"total":6,"successful":6,"skipped":0,"failed":0},"hits":{"total":{"value":708,"relation":"eq"},"max_score":1.0,"hits":[{"
_index":"wazuh-alerts-4.x-2025.12.14","_id":"7_UtH5sB1ByzhC3dEax7","_score":1.0,"_source":{"predecoder":{"hostname":"siem","program_name":"freshclam","timestamp
":"Dec 14 23:23:14"},"agent":{"name":"siem","id":"000"},"manager":{"name":"siem"},"rule":{"firedtimes":1,"mail":false,"level":3,"pci_dss":["5.2"],"tsc":["A1.2"]
,"description":"ClamAV database update","groups":["clamd","freshclam","virus"],"id":"52507","nist_800_53":["SI.3"],"gpg13":["4.4"],"gdpr":["IV_35.7.d"]},"decode
r":{"name":"freshclam"},"full_log":"Dec 14 23:23:14 siem freshclam[701]: Mon Dec 15 00:23:14 2025 -> ClamAV update process started at Mon Dec 15 00:23:14 2025",
"input":{"type":"log"},"@timestamp":"2025-12-14T23:23:15.144Z","location":"journald","id":"1765754595.407","timestamp":"2025-12-15T00:23:15.144+0100"}},{"_index
":"wazuh-alerts-4.x-2025.12.14","_id":"1_X1HpsB1ByzhC3d-6wp","_score":1.0,"_source":{"predecoder":{"hostname":"siem","program_name":"freshclam","timestamp":"Dec
 14 22:23:09"},"agent":{"name":"siem","id":"000"},"manager":{"name":"siem"},"rule":{"firedtimes":1,"mail":false,"level":3,"pci_dss":["5.2"],"tsc":["A1.2"],"desc
ription":"ClamAV database update","groups":["clamd","freshclam","virus"],"id":"52507","nist_800_53":["SI.3"],"gpg13":["4.4"],"gdpr":["IV_35.7.d"]},"decoder":{"n
ame":"freshclam"},"full_log":"Dec 14 22:23:09 siem freshclam[701]: Sun Dec 14 23:23:09 2025 -> ClamAV update process started at Sun Dec 14 23:23:09 2025","input
":{"type":"log"},"@timestamp":"2025-12-14T22:23:10.243Z","location":"journald","id":"1765750990.851264","timestamp":"2025-12-14T23:23:10.243+0100"}},{"_index":"
wazuh-alerts-4.x-2025.12.14","_id":"v_W_HpsB1ByzhC3dC6zP","_score":1.0,"_source":{"predecoder":{"hostname":"siem","program_name":"freshclam","timestamp":"Dec 14
 21:23:04"},"agent":{"name":"siem","id":"000"},"manager":{"name":"siem"},"rule":{"firedtimes":1,"mail":false,"level":3,"pci_dss":["5.2"],"tsc":["A1.2"],"descrip
tion":"ClamAV database update","groups":["clamd","freshclam","virus"],"id":"52507","nist_800_53":["SI.3"],"gpg13":["4.4"],"gdpr":["IV_35.7.d"]},"decoder":{"name
":"freshclam"},"full_log":"Dec 14 21:23:04 siem freshclam[701]: Sun Dec 14 22:23:04 2025 -> ClamAV update process started at Sun Dec 14 22:23:04 2025","input":{
"type":"log"},"@timestamp":"2025-12-14T21:23:05.114Z","location":"journald","id":"1765747385.850938","timestamp":"2025-12-14T22:23:05.114+0100"}},{"_index":"waz
uh-alerts-4.x-2025.12.14","_id":"nvV4HpsB1ByzhC3dl6yE","_score":1.0,"_source":{"agent":{"name":"siem","id":"000"},"previous_output":"Previous output:\nossec: ou
tput: 'netstat listening ports':\ntcp 0.0.0.0:22 0.0.0.0:* /usr\ntcp 0.0.0.0:443 0.0.0.0:* 2077/node\ntcp 0.0.0.0:1514 0.0.0.0:* 3060/wazuh-remoted\ntcp 0.0.0.0
:1515 0.0.0.0:* 2910/wazuh-authd\ntcp 127.0.0.1:9200 0.0.0.0:* 15222/java\ntcp 127.0.0.1:9300 0.0.0.0:* 15222/java\ntcp 0.0.0.0:55000 0.0.0.0:* 2861/python3","m
anager":{"name":"siem"},"rule":{"firedtimes":2,"mail":false,"level":7,"pci_dss":["10.2.7","10.6.1"],"hipaa":["164.312.b"],"tsc":["CC6.8","CC7.2","CC7.3"],"descr
iption":"Listened ports status (netstat) changed (new port opened or closed).","groups":["ossec"],"id":"533","nist_800_53":["AU.14","AU.6"],"gpg13":["10.1"],"gd
pr":["IV_35.7.d"]},"decoder":{"name":"ossec"},"full_log":"ossec: output: 'netstat listening ports':\ntcp 0.0.0.0:22 0.0.0.0:* /usr\ntcp 0.0.0.0:443 0.0.0.0:* 42
37/node\ntcp 0.0.0.0:1514 0.0.0.0:* 3060/wazuh-remoted\ntcp 0.0.0.0:1515 0.0.0.0:* 2910/wazuh-authd\ntcp 127.0.0.1:9200 0.0.0.0:* 15222/java\ntcp 127.0.0.1:9300
 0.0.0.0:* 15222/java\ntcp 0.0.0.0:55000 0.0.0.0:* 2861/python3","previous_log":"ossec: output: 'netstat listening ports':\ntcp 0.0.0.0:22 0.0.0.0:* /usr\ntcp 0
.0.0.0:443 0.0.0.0:* 2077/node\ntcp 0.0.0.0:1514 0.0.0.0:* 3060/wazuh-remoted\ntcp 0.0.0.0:1515 0.0.0.0:* 2910/wazuh-authd\ntcp 127.0.0.1:9200 0.0.0.0:* 15222/j
ava\ntcp 127.0.0.1:9300 0.0.0.0:* 15222/java\ntcp 0.0.0.0:55000 0.0.0.0:* 2861/python3","input":{"type":"log"},"@timestamp":"2025-12-14T20:06:16.034Z","location
":"netstat listening ports","id":"1765742776.849276","timestamp":"2025-12-14T21:06:16.034+0100"}},{"_index":"wazuh-alerts-4.x-2025.12.14","_id":"l_V0HpsB1ByzhC3
dF6wN","_score":1.0,"_source":{"agent":{"name":"siem","id":"000"},"manager":{"name":"siem"},"data":{"dpkg_status":"status installed","package":"gawk","arch":"am
d64","version":"1:5.2.1-2"},"rule":{"firedtimes":1,"mail":false,"level":7,"pci_dss":["10.6.1","10.2.7"],"hipaa":["164.312.b"],"tsc":["CC7.2","CC7.3","CC6.8","CC
8.1"],"description":"New dpkg (Debian Package) installed.","groups":["syslog","dpkg","config_changed"],"id":"2902","nist_800_53":["AU.6","AU.14"],"gpg13":["4.10
"],"gdpr":["IV_35.7.d"]},"decoder":{"name":"dpkg-decoder"},"full_log":"2025-12-14 21:01:11 status installed gawk:amd64 1:5.2.1-2","input":{"type":"log"},"@times
tamp":"2025-12-14T20:01:11.747Z","location":"/var/log/dpkg.log","id":"1765742471.847097","timestamp":"2025-12-14T21:01:11.747+0100"}}]}}root@siem:~# _
```


C’est bien le cas aussi, on part sur une bonne base.

Activation de l’archivage des logs (critique pour l’investigation)

Par défaut, Wazuh ne stocke que les alertes. Pour faire du threat hunting ou de la rétro-analyse, on a besoin de tous les logs bruts.

nano /var/ossec/etc/ossec.conf

Avant :

```
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>no</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>15m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
    <update_check>yes</update_check>
  </global>
```

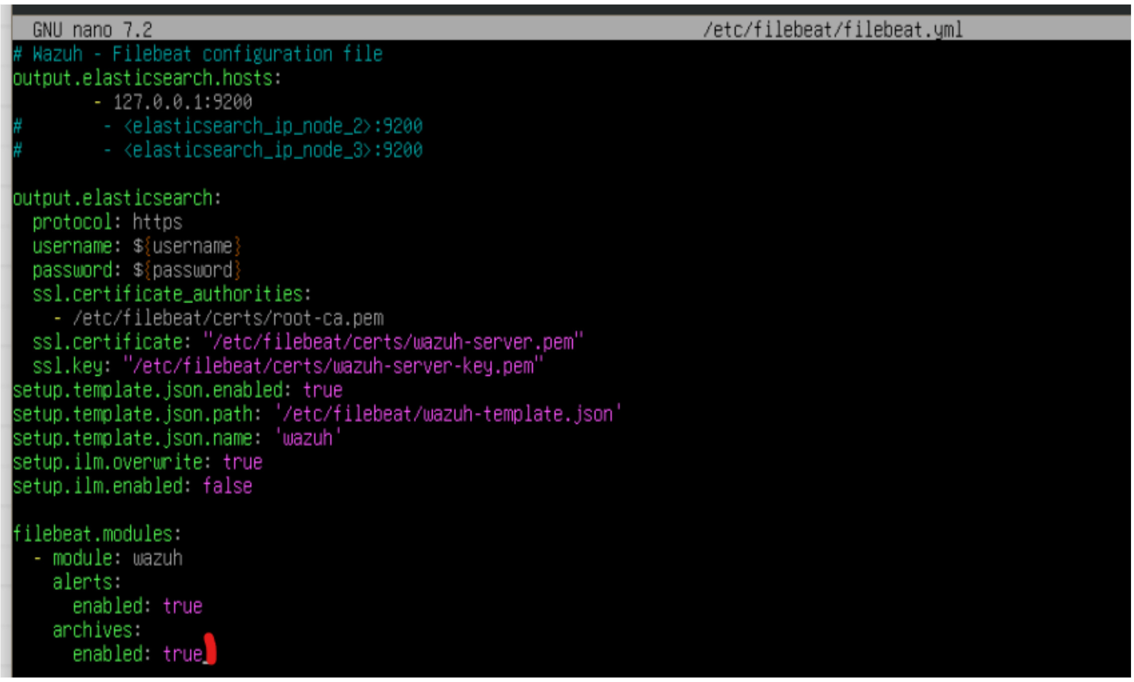
Après :

```
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>yes</logall>
    <logall_json>yes</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>15m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
    <update_check>yes</update_check>
  </global>
```

• <logall> active ou désactive l'archivage de tous les messages du journal. Lorsqu'il est activé, le serveur Wazuh stocke les journaux au format syslog.
• <logall_json> active ou désactive la journalisation des événements. Lorsqu'il est activé, le serveur Wazuh stocke les événements au format JSON.

systemctl restart wazuh-manager

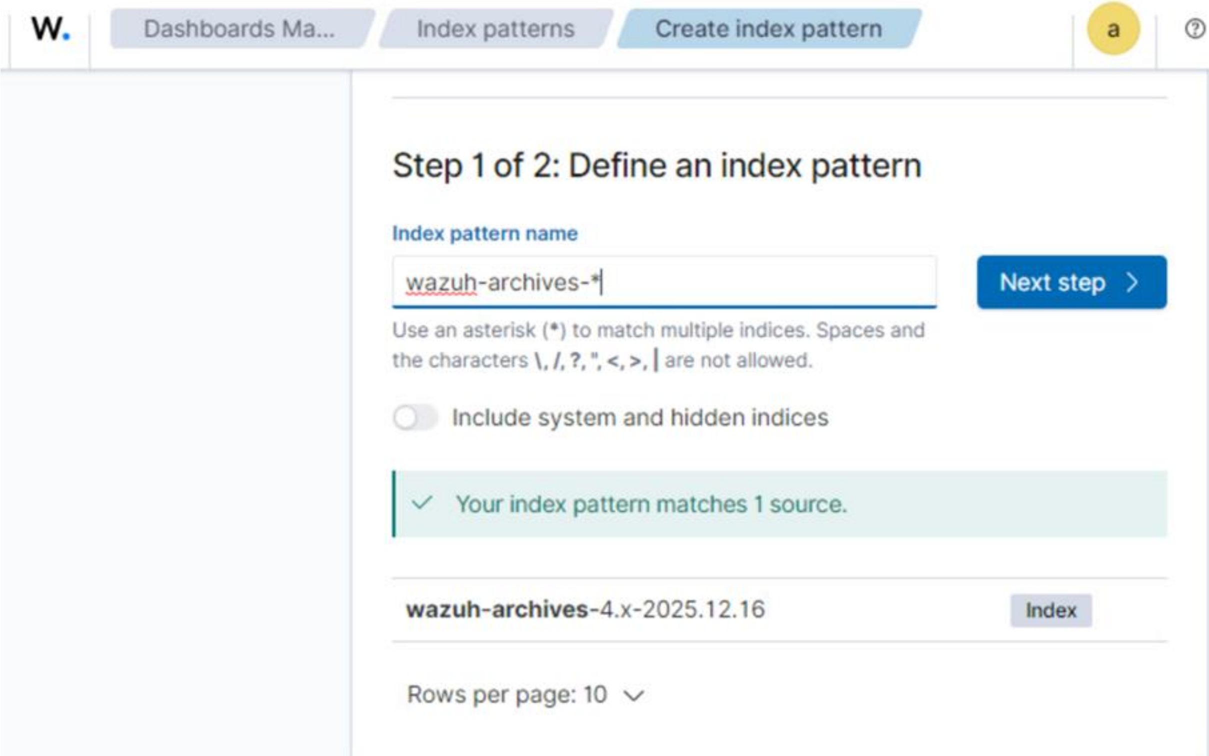
Visualisation des events sur le dashboard (archives en true) :

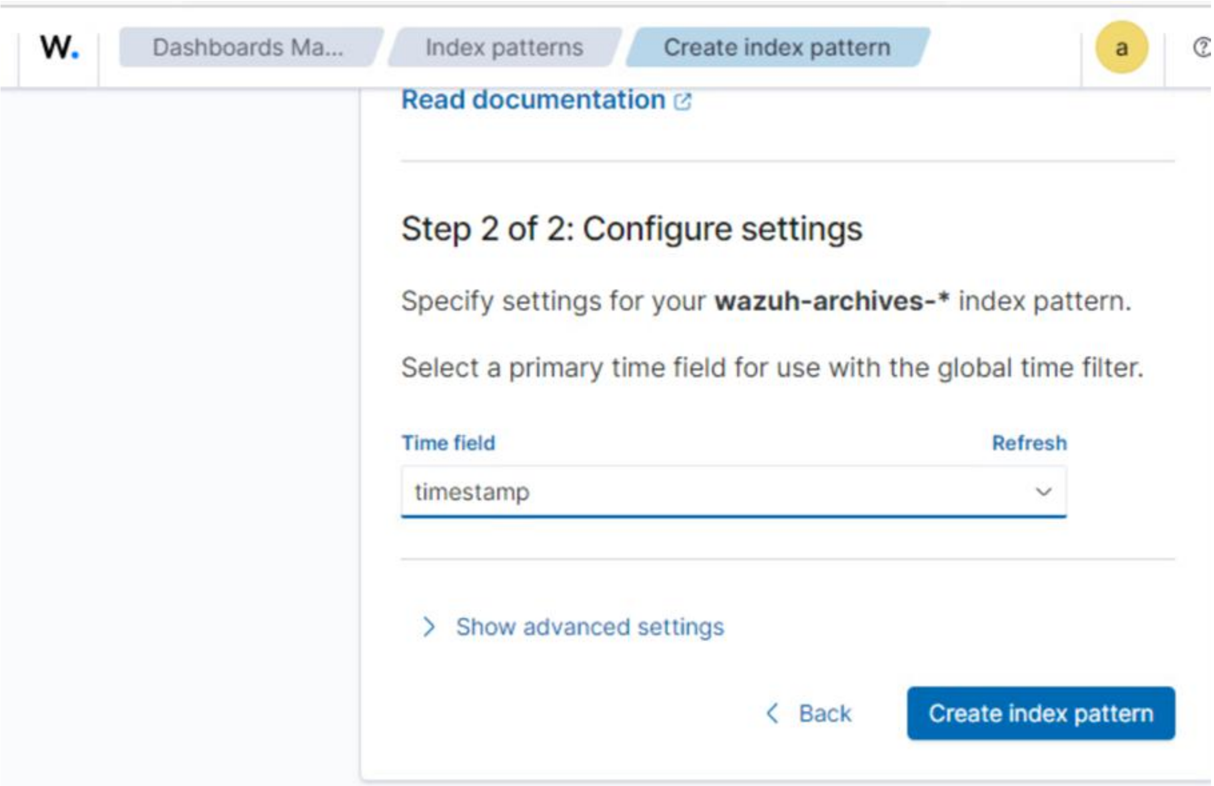


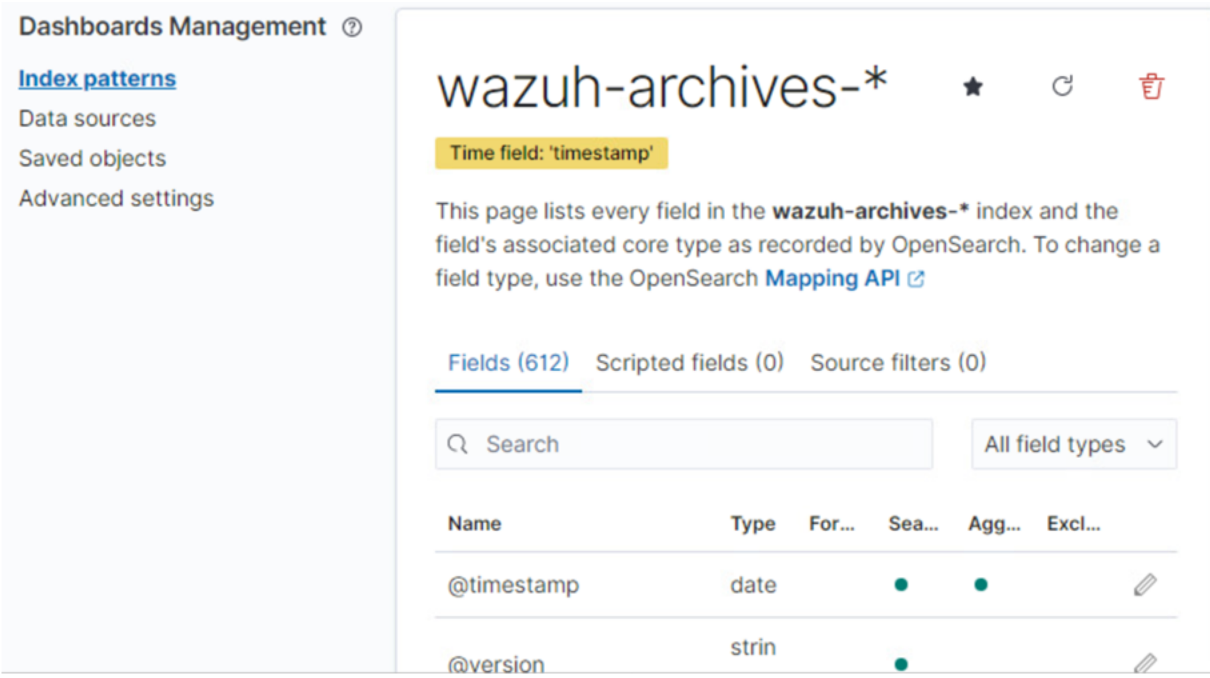
```
  GNU nano 7.2                                            /etc/filebeat/filebeat.yml
# Wazuh - Filebeat configuration file
output.elasticsearch.hosts:
        - 127.0.0.1:9200
#         - <elasticsearch_ip_node_2>:9200
#         - <elasticsearch_ip_node_3>:9200

output.elasticsearch:
  protocol: https
  username: ${username}
  password: ${password}
  ssl.certificate_authorities:
    - /etc/filebeat/certs/root-ca.pem
  ssl.certificate: "/etc/filebeat/certs/wazuh-server.pem"
  ssl.key: "/etc/filebeat/certs/wazuh-server-key.pem"
setup.template.json.enabled: true
setup.template.json.path: '/etc/filebeat/wazuh-template.json'
setup.template.json.name: 'wazuh'
setup.ilm.overwrite: true
setup.ilm.enabled: false

filebeat.modules:
  - module: wazuh
    alerts:
      enabled: true
    archives:
      enabled: true
```

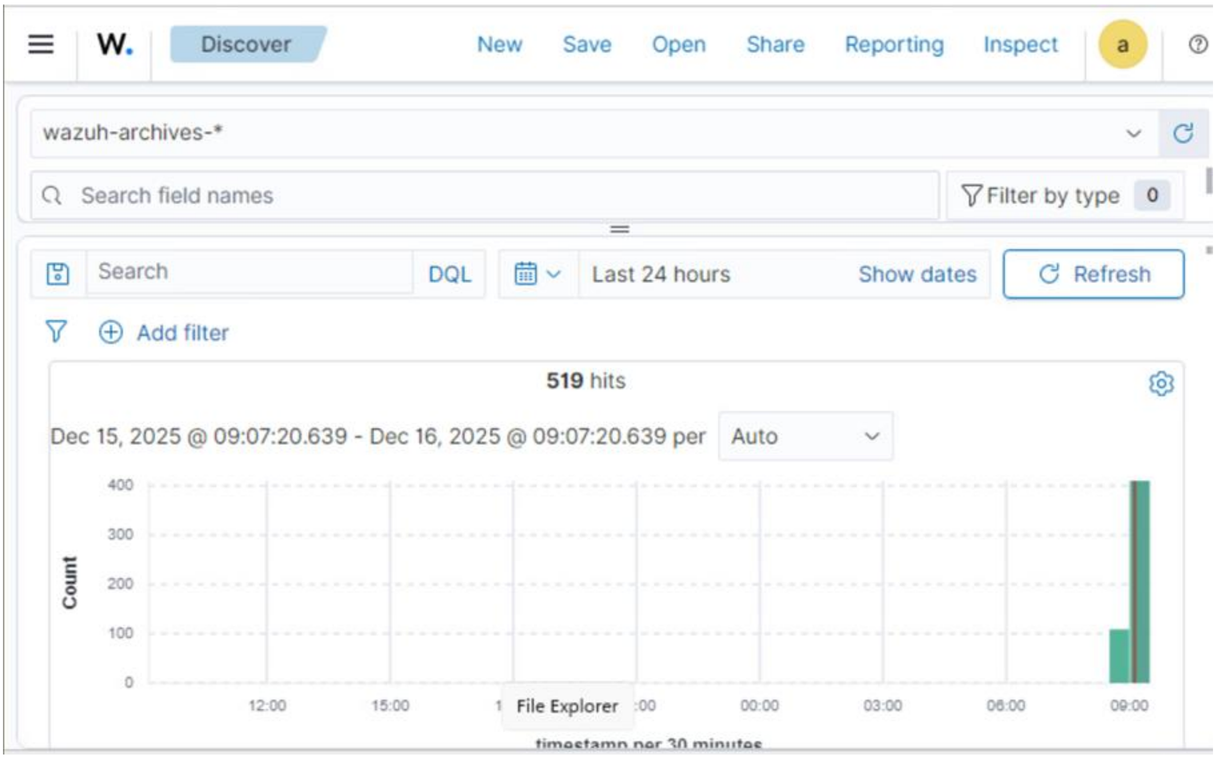
systemctl restart filebeat

Vérification de l’archivage :


W.
Dashboards Ma...
Index patterns
Create index pattern
a
Step 1 of 2: Define an index pattern
Index pattern name
wazuh-archives-*
Next step
Use an asterisk (*) to match multiple indices. Spaces and the characters \, /, ?, ", <, >, | are not allowed.
Include system and hidden indices
Your index pattern matches 1 source.
wazuh-archives-4.x-2025.12.16
Index
Rows per page: 10

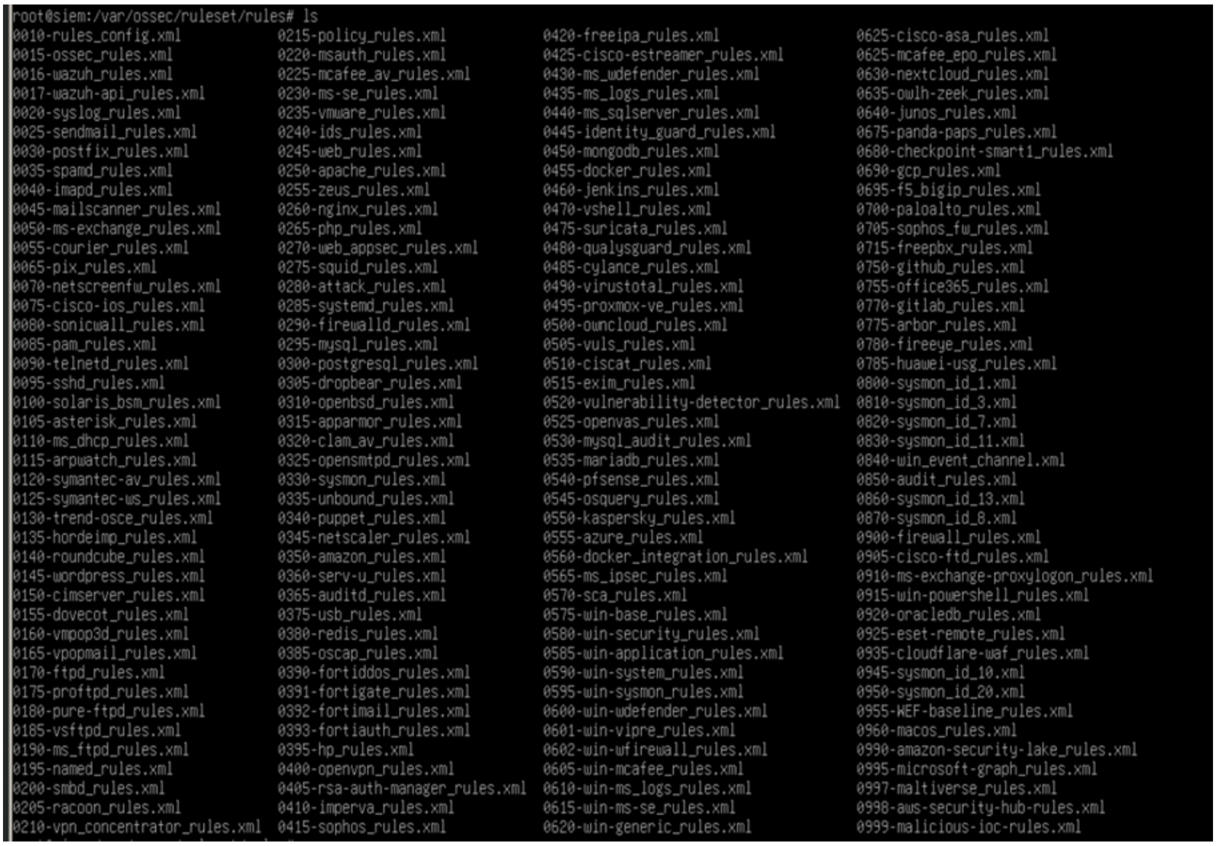

W.
Dashboards Ma...
Index patterns
Create index pattern
a
Read documentation
Step 2 of 2: Configure settings
Specify settings for your wazuh-archives-* index pattern.
Select a primary time field for use with the global time filter.
Time field
Refresh
timestamp
Show advanced settings
Back
Create index pattern





C’est tout bon là aussi

Pour les règles Wazuh, voici celles par défauts :

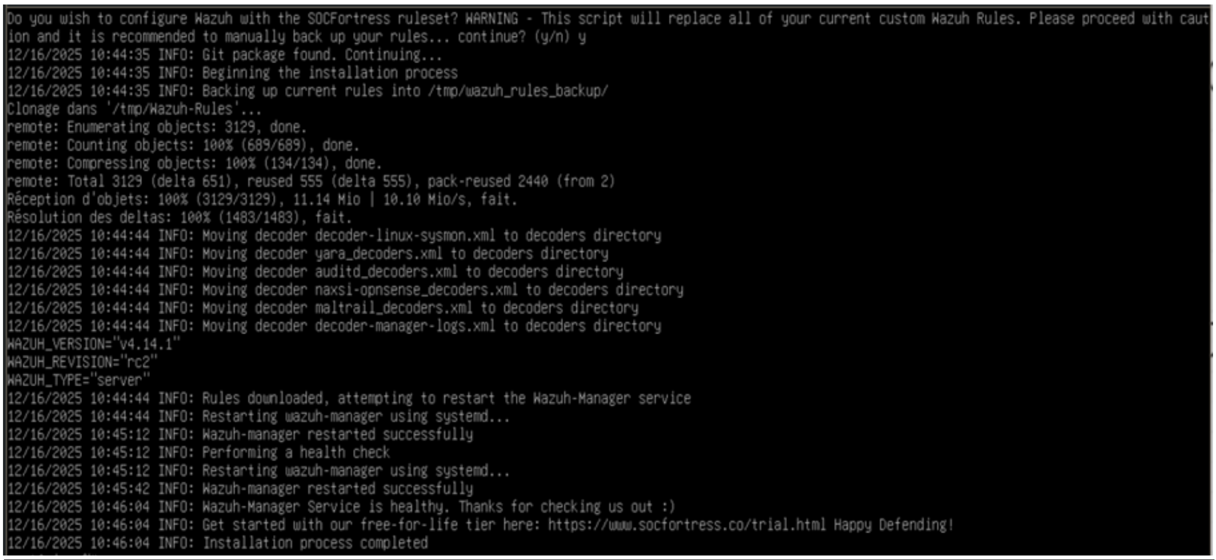


```
root@siem:/var/ossec/ruleset/rules# ls
0010-rules_config.xml          0215-policy_rules.xml            0420-freeipa_rules.xml                0625-cisco-asa_rules.xml
0015-ossec_rules.xml           0220-msauth_rules.xml            0425-cisco-estreamer_rules.xml        0625-mcafee_epo_rules.xml
0016-wazuh_rules.xml           0225-mcafee_av_rules.xml         0430-ms_wdefender_rules.xml           0630-nextcloud_rules.xml
0017-wazuh-api_rules.xml       0230-ms-se_rules.xml             0435-ms_logs_rules.xml                0635-owlh-zeek_rules.xml
0020-syslog_rules.xml          0235-vmware_rules.xml            0440-ms_sqlserver_rules.xml           0640-junos_rules.xml
0025-sendmail_rules.xml        0240-ids_rules.xml               0445-identity_guard_rules.xml         0675-panda-paps_rules.xml
0030-postfix_rules.xml         0245-web_rules.xml               0450-mongodb_rules.xml                0680-checkpoint-smart1_rules.xml
0035-spamd_rules.xml           0250-apache_rules.xml            0455-docker_rules.xml                 0690-gcp_rules.xml
0040-imapd_rules.xml           0255-zeus_rules.xml              0460-jenkins_rules.xml                0695-f5_bigip_rules.xml
0045-mailscanner_rules.xml     0260-nginx_rules.xml             0470-vshell_rules.xml                 0700-paloalto_rules.xml
0050-ms-exchange_rules.xml     0265-php_rules.xml               0475-suricata_rules.xml               0705-sophos_fw_rules.xml
0055-courier_rules.xml         0270-web_appsec_rules.xml        0480-qualysguard_rules.xml            0715-freepbx_rules.xml
0065-pix_rules.xml             0275-squid_rules.xml             0485-cylance_rules.xml                0750-github_rules.xml
0070-netscreenfw_rules.xml     0280-attack_rules.xml            0490-virustotal_rules.xml             0755-office365_rules.xml
0075-cisco-ios_rules.xml       0285-systemd_rules.xml           0495-proxmox-ve_rules.xml             0770-gitlab_rules.xml
0080-sonicwall_rules.xml       0290-firewalld_rules.xml         0500-owncloud_rules.xml               0775-arbor_rules.xml
0085-pam_rules.xml             0295-mysql_rules.xml             0505-vuls_rules.xml                   0780-fireeye_rules.xml
0090-telnetd_rules.xml         0300-postgresql_rules.xml        0510-ciscat_rules.xml                 0785-huawei-usg_rules.xml
0095-sshd_rules.xml            0305-dropbear_rules.xml          0515-exim_rules.xml                   0800-sysmon_id_1.xml
0100-solaris_bsm_rules.xml     0310-openbsd_rules.xml           0520-vulnerability-detector_rules.xml 0810-sysmon_id_3.xml
0105-asterisk_rules.xml        0315-apparmor_rules.xml          0525-openvas_rules.xml                0820-sysmon_id_7.xml
0110-ms_dhcp_rules.xml         0320-clam_av_rules.xml           0530-mysql_audit_rules.xml            0830-sysmon_id_11.xml
0115-arpwatch_rules.xml        0325-opensmtpd_rules.xml         0535-mariadb_rules.xml                0840-win_event_channel.xml
0120-symantec-av_rules.xml     0330-sysmon_rules.xml            0540-pfsense_rules.xml                0850-audit_rules.xml
0125-symantec-ws_rules.xml     0335-unbound_rules.xml           0545-osquery_rules.xml                0860-sysmon_id_13.xml
0130-trend-osce_rules.xml      0340-puppet_rules.xml            0550-kaspersky_rules.xml              0870-sysmon_id_8.xml
0135-hordeimp_rules.xml        0345-netscaler_rules.xml         0555-azure_rules.xml                  0900-firewall_rules.xml
0140-roundcube_rules.xml       0350-amazon_rules.xml            0560-docker_integration_rules.xml     0905-cisco-ftd_rules.xml
0145-wordpress_rules.xml       0360-serv-u_rules.xml            0565-ms_ipsec_rules.xml               0910-ms-exchange-proxylogon_rules.xml
0150-cimserver_rules.xml       0365-auditd_rules.xml            0570-sca_rules.xml                    0915-win-powershell_rules.xml
0155-dovecot_rules.xml         0375-usb_rules.xml               0575-win-base_rules.xml               0920-oracledb_rules.xml
0160-vmpop3d_rules.xml         0380-redis_rules.xml             0580-win-security_rules.xml           0925-eset-remote_rules.xml
0165-vpopmail_rules.xml        0385-oscap_rules.xml             0585-win-application_rules.xml        0935-cloudflare-waf_rules.xml
0170-ftpd_rules.xml            0390-fortiddos_rules.xml         0590-win-system_rules.xml             0945-sysmon_id_10.xml
0175-proftpd_rules.xml         0391-fortigate_rules.xml         0595-win-sysmon_rules.xml             0950-sysmon_id_20.xml
0180-pure-ftpd_rules.xml       0392-fortimail_rules.xml         0600-win-wdefender_rules.xml          0955-WEF-baseline_rules.xml
0185-vsftpd_rules.xml          0393-fortiauth_rules.xml         0601-win-vipre_rules.xml              0960-macos_rules.xml
0190-ms_ftpd_rules.xml         0395-hp_rules.xml                0602-win-wfirewall_rules.xml          0990-amazon-security-lake_rules.xml
0195-named_rules.xml           0400-openvpn_rules.xml           0605-win-mcafee_rules.xml             0995-microsoft-graph_rules.xml
0200-smbd_rules.xml            0405-rsa-auth-manager_rules.xml  0610-win-ms_logs_rules.xml            0997-maltiverse_rules.xml
0205-racoon_rules.xml          0410-imperva_rules.xml           0615-win-ms-se_rules.xml              0998-aws-security-hub-rules.xml
0210-vpn_concentrator_rules.xml 0415-sophos_rules.xml           0620-win-generic_rules.xml            0999-malicious-ioc-rules.xml
```

Nous pouvons rajouter des règles intéressantes, via ce projet github (https://github.com/socfortress/Wazuh-Rules?tab=readme-ov-file) :

Sur le serveur SIEM nous faisons ceci :

curl -so ~/wazuh_socfortress_rules.sh https://raw.githubusercontent.com/socfortress/Wazuh-Rules/main/wazuh_socfortress_rules.sh && bash ~/wazuh_socfortress_rules.sh



```
Do you wish to configure Wazuh with the SOCFortress ruleset? WARNING - This script will replace all of your current custom Wazuh Rules. Please proceed with caut
ion and it is recommended to manually back up your rules... continue? (y/n) y
12/16/2025 10:44:35 INFO: Git package found. Continuing...
12/16/2025 10:44:35 INFO: Beginning the installation process
12/16/2025 10:44:35 INFO: Backing up current rules into /tmp/wazuh_rules_backup/
Clonage dans '/tmp/Wazuh-Rules'...
remote: Enumerating objects: 3129, done.
remote: Counting objects: 100% (689/689), done.
remote: Compressing objects: 100% (134/134), done.
remote: Total 3129 (delta 651), reused 555 (delta 555), pack-reused 2440 (from 2)
Réception d'objets: 100% (3129/3129), 11.14 Mio | 10.10 Mio/s, fait.
Résolution des deltas: 100% (1483/1483), fait.
12/16/2025 10:44:44 INFO: Moving decoder decoder-linux-sysmon.xml to decoders directory
12/16/2025 10:44:44 INFO: Moving decoder yara_decoders.xml to decoders directory
12/16/2025 10:44:44 INFO: Moving decoder auditd_decoders.xml to decoders directory
12/16/2025 10:44:44 INFO: Moving decoder naxsi-opnsense_decoders.xml to decoders directory
12/16/2025 10:44:44 INFO: Moving decoder maltrail_decoders.xml to decoders directory
12/16/2025 10:44:44 INFO: Moving decoder decoder-manager-logs.xml to decoders directory
WAZUH_VERSION="v4.14.1"
WAZUH_REVISION="rc2"
WAZUH_TYPE="server"
12/16/2025 10:44:44 INFO: Rules downloaded, attempting to restart the Wazuh-Manager service
12/16/2025 10:44:44 INFO: Restarting wazuh-manager using systemd...
12/16/2025 10:45:12 INFO: Wazuh-manager restarted successfully
12/16/2025 10:45:12 INFO: Performing a health check
12/16/2025 10:45:12 INFO: Restarting wazuh-manager using systemd...
12/16/2025 10:45:42 INFO: Wazuh-manager restarted successfully
12/16/2025 10:46:04 INFO: Wazuh-Manager Service is healthy. Thanks for checking us out :)
12/16/2025 10:46:04 INFO: Get started with our free-for-life tier here: https://www.socfortress.co/trial.html Happy Defending!
12/16/2025 10:46:04 INFO: Installation process completed
```

On a tout ceci en règles custom maintenant :

```
root@siem:/var/ossec/etc/rules# ls
100002-suricata.xml                          111101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT12.xml  200850-crowdstrike.xml
100030-amazon_aws_cloudwatch.xml             112101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT13.xml  200900-wazuh_inventory.xml
100050-win_autoruns_rules.xml                113101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT14.xml  200910-wazuh_sca.xml
100060-win_sigcheck_rules.xml                114101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT15.xml  200920-dnstwist.xml
100070-win_logonsessions_rules.xml           116101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT17.xml  200930-duo.xml
100080-alienvault.xml                        117101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT18.xml  200940-mimecast.xml
100099-Modsecurity.xml                       121101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT22.xml  200960-dfir_iris.xml
100100-MITRE_TECHNIQUES_FROM_SYSMON_EVENT1.xml  121201-MITRE_TECHNIQUES_FROM_SYSMON_EVENT6.xml  200970-phishing.xml
100535-win_powershell_rules.xml              200001-windows_chainsaw_rules.xml                200980-socfortress.xml
100610-domain_stats_rules.xml                200050-chainsaw_sigma_rules.xml                  200990-healthcheck.xml
100620-misp.xml                              200070-sysmon_reload.xml                         201000-pentest-tools.xml
100623-opencti.xml                           200100-yara_rules.xml                            201010-ad_inventory.xml
100630-maltrail.xml                          200110-auditd.xml                                201015-software.xml
100651-abuseipdb.xml                         200150-sysmon_for_linux_rules.xml                300001-win_sigma_rules_builtin.xml
100660-beelzebub.xml                         200200-osquery.xml                               300100-cisco_secure_endpoint.xml
101101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT2.xml  200300-packetbeat_rules.xml                   400200-open-audit.xml
102101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT3.xml  200360-docker_falco_rules.xml                 500010-manager_logs.xml
106101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT7.xml  200400-nmap-scan_rules.xml                    600000-active_response.xml
108000-office365.xml                         200450-f-secure-epp_rules.xml                    700000-tetragon.xml
109000-microsoft_defender.xml                200460-opensense.xml                             800100-socfortress_added.xml
109100-win_sysmon_new_events.xml             200500-sap.xml                                   900000-exclusion_rules.xml
109101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT10.xml  200700-sophos.xml                            96600-snyk_rules.xml
110101-MITRE_TECHNIQUES_FROM_SYSMON_EVENT11.xml  200800-trendmicro.xml                        local_rules.xml
```

On peut vérifier si les nouvelles règles ont bien été chargés :

```
root@siem:/var/ossec/etc/rules# grep -n "rules" /var/ossec/logs/ossec.log
21:2025/12/16 09:00:16 sca: INFO: Starting evaluation of policy: '/var/ossec/ruleset/sca/cis_debian12.yml'
24:2025/12/16 09:00:37 sca: INFO: Evaluation finished for policy '/var/ossec/ruleset/sca/cis_debian12.yml'
75:2025/12/16 09:57:04 wazuh-analysisd: INFO: Total rules enabled: '8451'
125:2025/12/16 09:57:09 sca: INFO: Loaded policy '/var/ossec/ruleset/sca/cis_debian12.yml'
133:2025/12/16 09:57:09 sca: INFO: Starting evaluation of policy: '/var/ossec/ruleset/sca/cis_debian12.yml'
154:2025/12/16 09:57:28 sca: INFO: Evaluation finished for policy '/var/ossec/ruleset/sca/cis_debian12.yml'
476:2025/12/16 10:45:05 wazuh-analysisd: INFO: Total rules enabled: '62698'
500:2025/12/16 10:45:09 sca: INFO: Loaded policy '/var/ossec/ruleset/sca/cis_debian12.yml'
512:2025/12/16 10:45:09 sca: INFO: Starting evaluation of policy: '/var/ossec/ruleset/sca/cis_debian12.yml'
849:2025/12/16 10:45:35 wazuh-analysisd: INFO: Total rules enabled: '62698'
874:2025/12/16 10:45:39 sca: INFO: Loaded policy '/var/ossec/ruleset/sca/cis_debian12.yml'
878:2025/12/16 10:45:39 sca: INFO: Starting evaluation of policy: '/var/ossec/ruleset/sca/cis_debian12.yml'
903:2025/12/16 10:45:57 sca: INFO: Evaluation finished for policy '/var/ossec/ruleset/sca/cis_debian12.yml'
```

C’est bien le cas

Pour des meilleures logs sur toutes nos machines, nous avons optés pour auditd (que nous installons et configurons donc sur toutes nos machines) :

```
root@irplatform:~# apt-get install auditd
```

Règles pour curl, wget et bash (intéressant pour les machines sans GUI car c’est la méthode pour récupérer des données depuis internet : IP, domaine…) :

```
  GNU nano 7.2                                   /etc/audit/rules.d/soc.rules
-a always,exit -F arch=b64 -S execve -F exe=/usr/bin/curl -k curl_exec
-a always,exit -F arch=b64 -S execve -F exe=/usr/bin/wget -k wget_exec
-a always,exit -F arch=b64 -S execve -F exe=/bin/bash -k bash_exec

-w /tmp -p wa -k tmp_write
-w /var/tmp -p wa -k tmp_write
-w /home -p wa -k home_write
```

Vérification que c’est bien actif :

```
root@irplatform:~# augenrules --load
No rules
enabled 1
failure 1
pid 14552
rate_limit 0
backlog_limit 8192
lost 0
backlog 1
backlog_wait_time 60000
backlog_wait_time_actual 0
enabled 1
failure 1
pid 14552
rate_limit 0
backlog_limit 8192
lost 0
backlog 1
backlog_wait_time 60000
backlog_wait_time_actual 0
enabled 1
failure 1
pid 14552
rate_limit 0
backlog_limit 8192
lost 0
backlog 1
backlog_wait_time 60000
backlog_wait_time_actual 0
root@irplatform:~# auditctl -l
-a always,exit -F arch=b64 -S execve -F exe=/usr/bin/curl -F key=curl_exec
-a always,exit -F arch=b64 -S execve -F exe=/usr/bin/wget -F key=wget_exec
-a always,exit -F arch=b64 -S execve -F exe=/bin/bash -F key=bash_exec
-w /tmp -p wa -k tmp_write
-w /var/tmp -p wa -k tmp_write
-w /home -p wa -k home_write
```

Création d’une règle personnalisé (dans local_rules) lié à l’utilisation de curl et wget :

```
<group name="audit,">
  <rule id="100500" level="10">
    <if_sid>80700</if_sid>
    <field name="audit.key">curl_exec|wget_exec</field>
    <description>Outil de téléchargement suspect détecté sur $(hostname) : $(audit.exe) $(audit.execve.a1)</description>
  </rule>
</group>
```

Test logs wazuh (curl example.com depuis un serveur sans GUI) :

t	data.audit.exe	/usr/bin/curl
t	data.audit.execve.a0	curl
t	data.audit.execve.a1	example.com

On a modifié aussi des informations du syscheck pour le FIM (dans le fichier ossec.conf) pour tous nos serveur Linux:

```
<directories check_all="yes" report_changes="yes" realtime="yes">/home</directories>
```

Pour DVWA nous avons fait la même chose en ajoutant le contrôle du site :

```
<directories check_all="yes" report_changes="yes" realtime="yes">/home,/var/www/html</directories>
```

Pour la Windows, surveillance de Downloads et Desktop :

```
    <directories check_all="yes" report_changes="yes" realtime="yes">C:\Users\*\Downloads,C:\Users\*
\Desktop</directories>
```

Voici d'autres règles que nous avons finalement ajoutés dans le local rules :

```
<group name="audit,">
  <rule id="100500" level="10">
    <if_sid>80700</if_sid>
    <field name="audit.key">curl_exec|wget_exec</field>
    <description>Outil de téléchargement suspect détecté sur $(hostname) : $(audit.exe) $(audit.execve.a1)</description>
  </rule>
</group>

<group name="syscheck,windows_custom,">
  <rule id="100501" level="9">
    <if_sid>554</if_sid>
    <field name="file">\\Downloads\\|\\Desktop\\</field>
    <description>FIM : Nouveau fichier détecté dans un dossier sensible (User Profile)</description>
  </rule>
</group>


<group name="sysmon,windows,syscheck_ext,">
  <rule id="100502" level="10">
    <if_group>sysmon_event1</if_group>
    <field name="win.eventdata.currentDirectory">Downloads|Desktop</field>
    <description>SOC Alert: Exécution d'un processus depuis un répertoire utilisateur (Suspect)</description>
  </rule>
</group>

<group name="windows,persistence,">
  <rule id="100503" level="9">
    <if_sid>60106</if_sid>
    <field name="win.system.eventID">^4698$</field>
    <description>SOC Alert: Nouvelle tache planifiée créée sur Windows</description>
  </rule>
</group>

<group name="linux,audit,persistence,">
  <rule id="100504" level="9">
  <if_sid>80792</if_sid>
  <description>Soc Alert: Modification suspecte de la persistence Linux (cron)</description>
  </rule>
</group>
```

100501 Surveiller le Bureau et les Téléchargements
100502 Pour les exécutions Sysmon (Downloads/Desktop)
100503 Pour les tâches planifiées Windows
100504 Pour la persistance Linux (Cron/Auditd)